

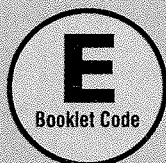
Roll No.:

Test Date: 22-12-2024

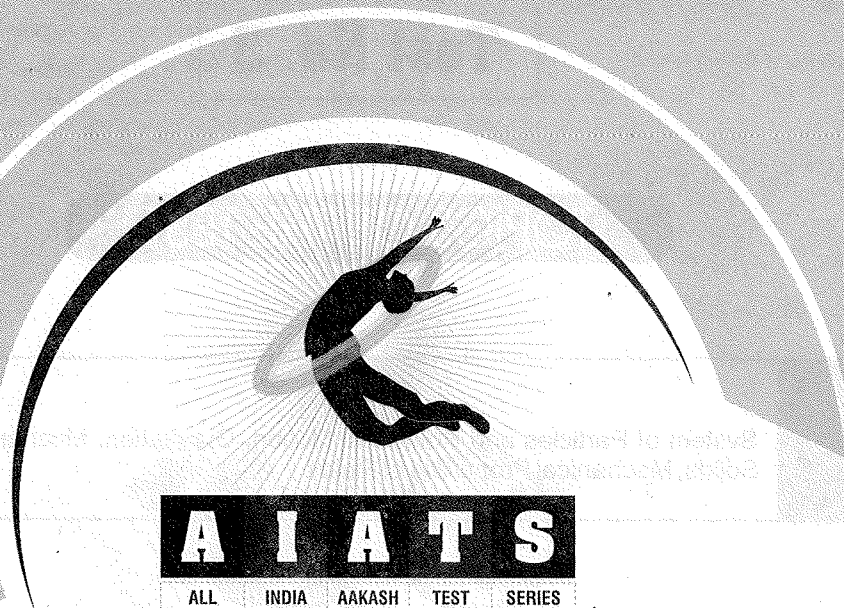


Aakash

Medical | IIT-JEE | Foundations



Aakash Delhi/NCR Zone



for

Medical Entrance Exam - 2025

National Eligibility-cum-Entrance Test (NEET)

TEST No. 3
(XII Passed Students)

INSTRUCTIONS FOR CANDIDATES

1. Read each question carefully.
2. It is mandatory to use Blue/Black Ball Point Pen to darken the appropriate circle in the answer sheet.
3. Mark should be dark and should completely fill the circle.
4. Rough work must not be done on the answer sheet.
5. Do not use white-fluid or any other rubbing material on answer sheet. No change in the answer once marked is allowed.
6. Student cannot use log tables and calculators or any other material in the examination hall.
7. Before attempting the question paper, student should ensure that the test paper contains all pages and no page is missing.
8. Each correct answer carries four marks. One mark will be deducted for each incorrect answer from the total score.
9. Before handing over the answer sheet to the invigilator, candidate should check that Roll No. and Centre Code have been filled and marked correctly.
10. Immediately after the prescribed examination time is over, the answer sheet to be returned to the invigilator.
11. There are two sections in each subject i.e., Section-A & Section-B. You have to attempt all 35 questions from Section-A & only 10 questions out of 15 from Section-B.

Note : It is compulsory to fill **Roll No.** and **Test Booklet Code** on answer sheet, otherwise your answer sheet will not be considered.

Test No. 3

TOPICS OF THE TEST

Physics

System of Particles and Rotational Motion, Gravitation, Mechanical Properties of Solids, Mechanical Properties of Fluids

Chemistry

Thermodynamics, Equilibrium, Redox Reactions, Electrochemistry

Botany

Anatomy of Flowering Plants, Plant Kingdom

Zoology

Excretory Products & their Elimination, Locomotion & Movement, Neural Control & Coordination



MM : 720

TEST - 3

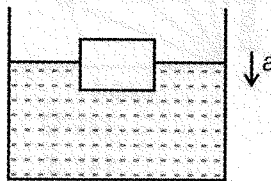
Time : 3 Hrs. 20 Min.

[PHYSICS]

Choose the correct answer :

SECTION-A

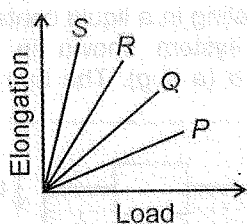
- Two masses m_1 and m_2 are d distance apart, where $m_1 > m_2$. The centre of mass of the system will
 - Lie at a distance $\frac{d}{2}$ from m_1
 - Be closer to m_2
 - Be closer to m_1
 - Lie at distance d from m_2
- The moment of inertia of a uniform semicircular ring of mass M and radius R about an axis perpendicular to plane of the ring and passing through its centre is
 - MR^2
 - $\frac{MR^2}{2}$
 - $2MR^2$
 - $\frac{3}{4}MR^2$
- A particle is moving in a circle with uniform speed. Its angular momentum about centre of the circle is constant due to
 - Constant non-zero torque
 - Constant force
 - Constant linear momentum
 - Zero torque
- An artificial satellite is revolving around earth in an elliptical orbital. Its
 - Kinetic energy is constant
 - Potential energy is constant
 - Total energy is constant
 - All of these
- Time period of a simple pendulum is given by $T = 2\pi\sqrt{\frac{l}{g}}$ in a lab, where l is length and g is acceleration due to gravity. If this pendulum is taken in a satellite revolving around earth, its time period
 - Will become half
 - Will become twice
 - Will remain unchanged
 - Will become infinite
- The escape speed (v_e) of an object of mass m from the surface of earth of mass M and radius R is such that
 - $v_e \propto m$
 - $v_e \propto \sqrt{m}$
 - $v_e \propto \sqrt{M}$
 - $v_e \propto M$
- Time period of revolution of a satellite around a planet is 4 hours. If the radius of orbit of the satellite is increased to 4 times of its previous value, the new time period of revolution will be
 - 8 hours
 - 16 hours
 - 32 hours
 - 64 hours
- A body is floating in a liquid contained in a beaker. The whole system shown in figure falls with acceleration a ($a < g$). The buoyant force on the body will



Space for Rough Work



- (1) Decrease
(2) Increase
(3) Remain unchanged
(4) May increase or decrease
9. Work done in increasing the radius of a soap bubble from 2 cm to 4 cm is nearly (surface tension of soap solution is 0.03 N m^{-1})
(1) $0.3\pi \text{ mJ}$ (2) $0.6\pi \text{ mJ}$
(3) $1.2\pi \text{ mJ}$ (4) $12\pi \text{ mJ}$
10. The Young's modulus of a wire of length L and area of cross-section A is 'Y'. If the length of wire is stretched to $2L$ and area is reduced to $\frac{A}{2}$, then its Young's modulus will
(1) Be $\frac{Y}{2}$ (2) Be $2Y$
(3) Remain Y (4) Be $4Y$
11. The Bulk modulus of a spherical object is B . If it is subjected to a uniform pressure P from all sides, the fractional decrease in its area will be
(1) $\frac{P}{B}$ (2) $\frac{P}{2B}$
(3) $\frac{P}{3B}$ (4) $\frac{2P}{3B}$
12. Four wires P , Q , R and S are made of same material and same length with different radii. The graphs in the figure shows the elongation-load variation. The thinnest wire is

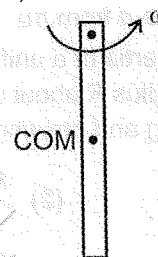


- (1) P (2) Q
(3) R (4) S

13. Two identical particles are moving with speed v and $\sqrt{3}v$ in perpendicular directions. The speed of their centre of mass is

- (1) $2v$ (2) v
(3) $\frac{\sqrt{3}v}{2}$ (4) $2\sqrt{3}v$

14. A thin uniform rod of mass m and length l is swinging freely about a horizontal axis passing through its one end. When it is at bottom-most position as shown in figure its angular velocity is ω . If centre of mass of the rod can rise to a maximum height $\frac{l}{4}$ from this position, then ω is



- (1) $\sqrt{\frac{3g}{l}}$ (2) $\sqrt{\frac{3g}{2l}}$
(3) $\sqrt{\frac{2g}{3l}}$ (4) $\frac{1}{2}\sqrt{\frac{3g}{l}}$

15. A thin horizontal circular disc is rotating about a vertical axis passing through its centre and perpendicular to its plane. An insect is at rest at the centre of the disc initially. The insect then moves along a radius to reach the rim of the disc. During the journey of the insect, the angular speed of the disc

- (1) Continuously decreases
(2) Continuously increases
(3) First increases and then decreases
(4) Remains unchanged

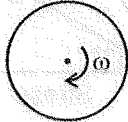
Space for Rough Work



16. Moment of inertia of uniform rod about an axis passing through mid-point and perpendicular to rod is 2400 kg m^2 . The moment of inertia about an axis passing through one of its ends and perpendicular to the rod is

(1) 600 kg m^2 (2) 1200 kg m^2
(3) 4800 kg m^2 (4) 9600 kg m^2

17. A disc is rotating about its axis as shown in the figure below. Its angular speed is ω . If M is the mass of disc and R is its radius, then kinetic energy of rotation of the disc is



(1) $\frac{MR^2\omega^2}{2}$ (2) $\frac{MR^2\omega^2}{4}$
(3) $\frac{3}{2}MR^2\omega^2$ (4) $MR^2\omega^2$

18. **Statement-I:** Under the action of a constant couple, angular acceleration of a rod is constant.

Statement-II: Under the action of a couple, acceleration of centre of mass of a rod is zero.

Select the correct option.

(1) Both statements I and II are correct
(2) Both statements I and II are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

19. A rigid body is rotating about a fixed axis with angular velocity ω . Its moment of inertia is I and kinetic energy is K . Its angular momentum about the same axis is

(1) $\sqrt{2KI}$ (2) $\frac{2K}{\omega}$
(3) $2K\omega$ (4) Both (1) and (2)

20. If g is the gravitational field intensity on the surface of earth, then value of gravitational field intensity at a height $\frac{R}{2}$ (R is the radius of earth) above the earth's surface is

(1) $\frac{g}{2}$ (2) $\frac{2}{9}g$
(3) $\frac{4}{9}g$ (4) $\frac{4}{7}g$

21. A satellite is revolving around earth at height R above the earth's surface. If M is the mass of earth, R is the radius of earth and m is the mass of the satellite, then its total energy is

(1) $-\frac{GMm}{2R}$ (2) $-\frac{GMm}{4R}$
(3) $-\frac{GMm}{R}$ (4) $-\frac{3GMm}{4R}$

22. A smooth tunnel is dug along the diameter of earth and a particle is released from the surface in the tunnel. If M is the mass of earth and R is its radius, then velocity of the particle at the centre is

(1) $\sqrt{\frac{GM}{R}}$ (2) $\sqrt{\frac{GM}{2R}}$
(3) $\sqrt{\frac{GM}{3R}}$ (4) $\sqrt{\frac{2GM}{3R}}$

23. The gravitational force between two masses M_1 and M_2 is F when placed distance d apart. If both the masses are doubled and the distance between them is reduced to half, the force between them

(1) Becomes $4F$ (2) Becomes $8F$
(3) Remains same (4) Becomes $16F$

24. The gravitational field of earth is
a. Zero at centre
b. Zero at infinite distance from centre of earth
c. Maximum at surface

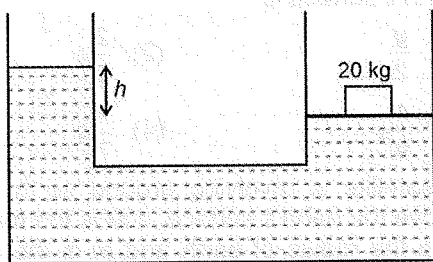
The correct statement(s) is/are

(1) a only (2) a and b only
(3) a and c only (4) All a, b and c

Space for Rough Work



25. The area of cross-section of the wider tube shown in the figure is 400 cm^2 . If a mass of 20 kg is placed on the massless piston, then the difference in height h of the level of water in the two tubes will be

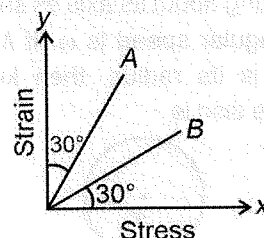


- (1) 20 cm (2) 50 cm
(3) 25 cm (4) 10 cm
26. A cylindrical tube of spray pump has radius 5 cm , one end of which has 20 fine holes of radius 1 mm . If the speed of liquid in the tube is 0.5 cm s^{-1} , then speed of water jet through the holes will be
- (1) 33.3 cm s^{-1} (2) 37.5 cm s^{-1}
(3) 42.5 cm s^{-1} (4) 62.5 cm s^{-1}
27. Equal volumes of water and a liquid of relative density 2 are mixed together, then the mixture has the density of
- (1) $\frac{2}{3} \text{ g cm}^{-3}$ (2) $\frac{3}{2} \text{ g cm}^{-3}$
(3) $\frac{4}{3} \text{ g cm}^{-3}$ (4) $\frac{3}{4} \text{ g cm}^{-3}$
28. A cubical block of mass 100 kg and density 600 kg m^{-3} floats in water. The volume of block outside water is
- (1) $\frac{1}{6} \text{ m}^3$ (2) $\frac{1}{4} \text{ m}^3$
(3) $\frac{1}{15} \text{ m}^3$ (4) $\frac{1}{24} \text{ m}^3$

29. Which of the following quantities does not have the unit of force per unit area?

(1) Young's modulus (2) Bulk modulus
(3) Breaking stress (4) Longitudinal strain

30. The strain-stress graph for wires A and B are as shown in the figure. If Y_A and Y_B are the Young's moduli of the materials, then



(1) $2Y_A = Y_B$ (2) $Y_A = Y_B$
(3) $Y_A = 3Y_B$ (4) $3Y_A = Y_B$

31. A wire can bear a maximum force F . Another wire of same material but twice the radius can bear the maximum force of

(1) F (2) $2F$
(3) $4F$ (4) $8F$

32. A planet P has density ρ and radius r and another planet Q has density 2ρ and radius $\frac{r}{2}$. The ratio of gravitational field intensities on their surfaces is

(1) $1 : 1$ (2) $1 : 2$
(3) $1 : 4$ (4) $1 : 8$

33. Gravitational potential of a uniform spherical shell of mass M and radius R on the surface is V . The potential at a distance $\frac{R}{2}$ from the centre of shell is

(1) $\frac{V}{2}$ (2) $\frac{V}{4}$
(3) V (4) $2V$

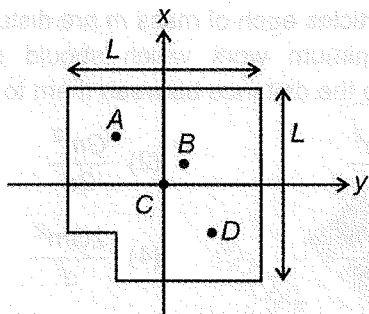
Space for Rough Work



34. Two masses m and $2m$ are held at distance d apart. When they are released, they move toward each other and collide. The distance moved by mass m just before collision is

- (1) $\frac{d}{3}$ (2) $\frac{d}{2}$
(3) $\frac{3}{4}d$ (4) $\frac{2}{3}d$

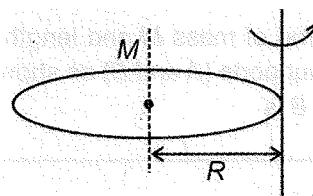
35. A uniform square sheet of side L is placed in xy plane with its centre at origin. Now a square section of size $\frac{L}{4}$ is removed from the sheet as shown. The point which is more likely to be COM of remaining sheet is



- (1) A (2) B
(3) C (4) D

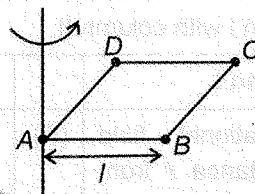
SECTION - B

36. A body of mass m is projected vertically upward with speed $\sqrt{\frac{GM}{R}}$ from the surface of earth. If M is the mass of earth and R is its radius, then maximum height attained by the body above earth's surface is
(1) $2R$ (2) R
(3) $\frac{R}{2}$ (4) $\frac{R}{4}$
37. Moment of inertia of a uniform disc of mass M and radius R about an axis passing through its rim and perpendicular to the plane of the disc as shown is MK^2 , where K is



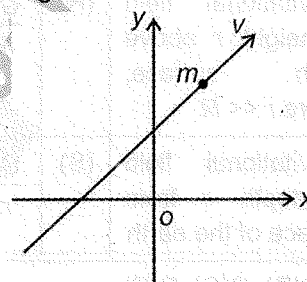
- (1) $\sqrt{\frac{3}{2}}R$ (2) $\sqrt{\frac{5}{2}}R$
(3) $\frac{\sqrt{3}R}{2}$ (4) $\frac{\sqrt{5}R}{2}$

38. Four point masses each of value m , are fixed at the corners of a square $ABCD$ of side l . The moment of inertia of this system about an axis passing through A and perpendicular to plane of the square is



- (1) $2ml^2$ (2) $3ml^2$
(3) $4ml^2$ (4) $5ml^2$

39. A particle of mass m is moving in x - y plane with constant velocity v as shown. Its angular momentum about the origin

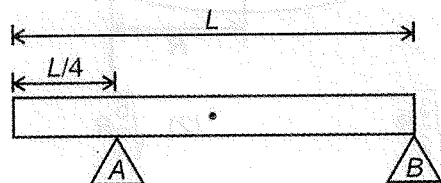


- (1) Remains constant
(2) Must increase
(3) Must decrease
(4) May increase or decrease

Space for Rough Work



40. A uniform rod of mass M and length L is balanced using two supports (A and B) as shown. The normal reaction at B is



- (1) $\frac{Mg}{3}$ (2) $\frac{Mg}{2}$
 (3) $\frac{2Mg}{3}$ (4) $\frac{Mg}{4}$
41. If M is the mass of earth and R is its radius, then match column I with column II.

	Column I		Column II
A.	Gravitational field at distance r from center, where $r < R$	(P)	$\frac{GM}{R^2} \left(1 - \frac{r}{R}\right)$
B.	Gravitation field at distance r from centre, where $r > R$	(Q)	$\frac{GM}{R^2} \left(1 - \frac{2r}{R}\right)$
C.	Gravitational field at height r above earth surface, where $r \ll R$	(R)	$\frac{GM}{R^3} r$
D.	Gravitational field at depth r from surface of the earth	(S)	$\frac{GM}{r^2}$

- (1) A(S), B(R), C(Q), D(P)
 (2) A(R), B(S), C(Q), D(P)
 (3) A(R), B(S), C(P), D(Q)
 (4) A(S), B(R), C(P), D(Q)

42. The correct expression for viscous drag acting on a metal sphere of diameter d , falling through a fluid of viscosity η with a velocity v will be

- (1) $6\pi\eta dv$
 (2) $3\pi\eta dv$
 (3) $\frac{3\pi\eta dv}{2}$
 (4) $4\pi\eta dv$

43. Which of the following is/are Kepler's laws?

- (1) Law of orbit (2) Law of areas
 (3) Law of continuity (4) Both (1) and (2)

44. Two particles each of mass m are distance d apart. The minimum work which should be done to increase the distance between them to $2d$ will be

- (1) $\frac{Gm^2}{d}$ (2) $\frac{Gm^2}{2d}$
 (3) $\frac{3Gm^2}{2d}$ (4) $\frac{2Gm^2}{d}$

45. A uniform rod of length L has mass M . If the area of cross-section is A and Young's modulus of rod is Y , then elongation in rod due to its own weight will be

- (1) $\frac{MgL}{AY}$ (2) $\frac{MgL}{3AY}$
 (3) $\frac{MgL}{2AY}$ (4) $\frac{2MgL}{AY}$

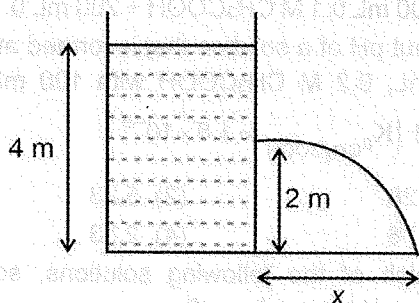
46. The correct dimensional formula for compressibility will be

- (1) $[MLT^{-2}]$
 (2) $[M^{-1}L^1T^2]$
 (3) $[ML^{-1}T^{-2}]$
 (4) $[ML^2T^2]$

Space for Rough Work



47. A drum is placed on the floor and it is filled with a liquid of almost no viscosity up to a height 4 m. A small opening is made at height 2 m from the base. The liquid jet strikes the floor at a distance x from the wall of the drum, then x is equal to



- (1) 4 m (2) 6 m
(3) 8 m (4) $\sqrt{40}$ m
48. A long capillary tube is dipped in water. The water rises to a height of 10 cm in it. If the entire arrangement is put in a lift which is accelerating upward with an acceleration 2 m s^{-2} , then height of liquid column in the capillary
- (1) Will be less than 10 cm
(2) Will be greater than 10 cm
(3) Will become 20 cm
(4) Will become zero

49. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): If no external force acts on a system of particles, the centre of mass will move with constant velocity or remain at rest.

Reason (R): Acceleration of centre of mass of system depends on external forces only.

In the light of above statements, select the correct option.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false
50. Choose the incorrect statement about gravitational force.
- (1) It forms action-reaction pair
(2) It is a central force
(3) It does not depend upon the nature of medium between the masses
(4) It is a non-conservative force

[CHEMISTRY]

SECTION - A

51. In which of the following reactions K_P and K_C will be equal?

- (1) $2\text{NH}_3(\text{g}) \rightleftharpoons \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$
(2) $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$
(3) $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons 2\text{HCl}(\text{g})$
(4) $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{H}_2\text{O}(\text{g})$

52. Number of H^+ ions present in 1 mL aqueous HNO_3 solution with $\text{pH} = 2.7$ is

[Given, $10^{-2.7} = 2 \times 10^{-3}$]

- (1) 1.204×10^{21} (2) 1.204×10^{18}
(3) 2.4×10^{19} (4) 6.02×10^{20}

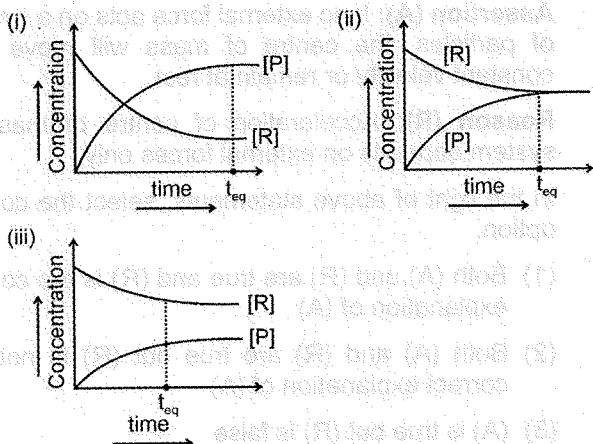
53. If K_w of pure water at $T^\circ\text{C}$ is 10^{-13} then its pH will be

- (1) 7 (2) 8
(3) 7.5 (4) 6.5

Space for Rough Work

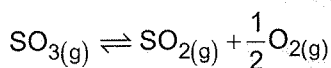


54. Consider the following graphs between concentration of reactant (R) and product (P) v/s time



Which of the above graphs is/are correctly represented?

- (1) (ii) only (2) (i) & (ii) only
(3) (i), (ii) & (iii) (4) (i) & (iii) only
55. If equilibrium constant K_c for reaction, $2\text{SO}_{2(g)} + \text{O}_{2(g)} \rightleftharpoons 2\text{SO}_{3(g)}$ is 9, then what will be equilibrium constant for following reaction at same temperature?



- (1) 3 (2) $\frac{1}{3}$
(3) 27 (4) $\frac{1}{\sqrt{3}}$
56. If equal volumes of two acids, with pH 2 and 3 respectively are mixed, then pH of resultant solution will nearly be
- (1) 5 (2) 1
(3) 2.5 (4) 2.26

57. Which of the following will form a buffer solution?

- (1) 100 mL 0.1 M CH_3COOH + 100 mL 0.1 M NaOH
(2) 100 mL 0.1 M CH_3COOH + 50 mL 0.1 M NaOH
(3) 100 mL 0.1 M CH_3COOH + 150 mL 0.1 M NaOH
(4) 100 mL 0.1 M CH_3COOH + 200 mL 0.1 M NaOH

58. Find out pH of a solution that is formed after mixing 100 mL, 0.2 M CH_3COOH with 100 mL, 0.05 M NaOH ($K_{a_{\text{CH}_3\text{COOH}}} = 1.8 \times 10^{-5}$)

- (1) 4.28 (2) 5.23
(3) 4.76 (4) 2.53

59. In which of the following solutions, solubility of BaSO_4 will be maximum?

- (1) 0.1 M BaCl_2 (2) 0.1 M Na_2SO_4
(3) 0.1 M NaCl (4) 0.01 M H_2SO_4

60. Work done for reversible expansion of 2 mol ideal gas from 100 mL to 1 L at constant 27°C temperature will be,

- (1) -2.76 cal
(2) -2.76 kcal
(3) +2.76 kcal
(4) +2.76 cal

61. Given below are two statements.

Statement I: Reversible adiabatic processes are isentropic in nature.

Statement II: Work done on the system during isothermal reversible process is stored as the internal energy of system.

In the light of above statements, choose the **correct** answer.

- (1) Statement I is true and statement II is false
(2) Both statement I and statement II are true
(3) Statement I is false and statement II is true
(4) Both statement I and statement II are false

Space for Rough Work



62. Match the processes in column I with quantitative value in column II for ideal gas:

	Column I		Column II
a.	Isothermal	(i)	$q = \Delta H$
b.	Isobaric	(ii)	$q = \Delta U$
c.	Adiabatic	(iii)	$w = \Delta U$
d.	Isochoric	(iv)	$q = -w$

Choose the correct answer.

- (1) a(iv), b(i), c(iii), d(ii) (2) a(i), b(iv), c(iii), d(ii)
 (3) a(iv), b(i), c(ii), d(iii) (4) a(iv), b(iii), c(i), d(ii)
63. If enthalpy of combustion for glucose is -2802 kJ/mol then amount of heat released on combustion of 6 g glucose will be
 (1) 2802 kJ (2) 93.4 kJ
 (3) 934 kJ (4) 280.2 kJ
64. Equilibrium constant for a reaction with standard Gibbs free energy of -9.2 kcal at 127°C will be
 (1) 10^{10} (2) 10^5
 (3) 50 (4) 500
65. If 2 A current is passed through 1 L brine solution for 965 second then which of the following is incorrect?
 (1) Total mole of gas evolved is 0.02
 (2) pH solution after electrolysis is 12.3
 (3) Total volume of gases evolved at STP = 0.224 L
 (4) Reduction of water will take place at cathode
66. Number of Faradays required to reduce 0.1 mol nitrobenzene to aniline will be,
 (1) 0.3 F (2) 0.9 F
 (3) 0.6 F (4) 0.1 F

67. Which among the following is true for adiabatic free expansion of an ideal gas?

- (1) $w < 0$ (2) $q > 0$
 (3) $\Delta S_{\text{total}} > 0$ (4) $\Delta U < 0$

68. Choose the correct statement(s) among the following

- a. If $Q_{\text{sp}} > K_{\text{sp}}$ then precipitation will take place.
 b. Ice melts faster at high pressure.
 c. For a sparingly soluble salt solubility is less than 0.01 M.

- (1) (a) & (b) only (2) (b) & (c) only
 (3) (a) & (c) only (4) (a), (b) & (c)

69. During reversible adiabatic expansion,

- (1) Entropy of system increases
 (2) Enthalpy of system increases
 (3) Temperature decreases
 (4) Internal energy remains constant

70. Which among the following is not a redox reaction?

- (1) $\text{PCl}_{5(g)} \rightarrow \text{PCl}_{3(g)} + \text{Cl}_{2(g)}$
 (2) $\text{CaCO}_{3(s)} \rightarrow \text{CaO}_{(s)} + \text{CO}_{2(g)}$
 (3) $2\text{H}_{2(g)} + \text{O}_{2(g)} \rightarrow 2\text{H}_2\text{O}_{(g)}$
 (4) $2\text{HI}_{(g)} \rightarrow \text{H}_{2(g)} + \text{I}_{2(g)}$

71. Which among the following metals will not give 'H₂' gas on reaction with dilute H₂SO₄?

- (1) Na (2) Ca
 (3) Pt (4) Al

72. If heat at constant volume for combustion of CH₄ at 127°C is -20 kJ then heat at constant pressure will be,

- (1) -14 kJ
 (2) -20 kJ
 (3) -28 kJ
 (4) -12 kJ

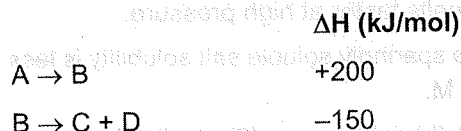
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73. Moles of acidic KMnO_4 required to oxidise 0.1 mol of FeC_2O_4 are,

- (1) 0.1
(2) 0.06
(3) 0.5
(4) 0.6

74. Consider the following processes



For $A \rightarrow C + D$; ΔH will be

- (1) -50 kJ (2) +50 kJ
(3) -350 kJ (4) +350 kJ

75. Solubility of Ba(OH)_2 in a solution with $\text{pH} = 12$ will be, $\{K_{sp} \text{ Ba(OH)}_2 = 10^{-12}\}$

- (1) 10^{-6} (2) 10^{-8}
(3) 10^{-10} (4) 10^{-9}

76. $a\text{Br}_2 + b\text{NaOH} \rightarrow c\text{NaBr} + d\text{NaBrO}_3 + e\text{H}_2\text{O}$.

Match the following according to above reaction.

	Column I		Column II
(i)	a	(P)	1
(ii)	b	(Q)	3
(iii)	c	(R)	6
(iv)	d	(S)	5

Correct answer is

- (1) (i) $\rightarrow$ (Q), (ii) $\rightarrow$ (R), (iii) $\rightarrow$ (S), (iv) $\rightarrow$ (P)
(2) (i) $\rightarrow$ (Q), (ii) $\rightarrow$ (R), (iii) $\rightarrow$ (P), (iv) $\rightarrow$ (S)
(3) (i) $\rightarrow$ (R), (ii) $\rightarrow$ (Q), (iii) $\rightarrow$ (S), (iv) $\rightarrow$ (P)
(4) (i) $\rightarrow$ (R), (ii) $\rightarrow$ (Q), (iii) $\rightarrow$ (P), (iv) $\rightarrow$ (S)

77. From the statements given below, correct statement(s) for reversible process is/are

- (a) The process could, at any moment, be reversed by an infinitesimal change.
(b) It proceeds very fast.
(c) System and surroundings are always in near equilibrium with each other.

- (1) (a), (b) & (c) (2) (a) & (b) only
(3) (a) & (c) only (4) (a) only

78. Intensive property among the following is

- (1) Volume (2) Number of mole
(3) Pressure (4) Surface area

79. Given below are two statements

Statement I: Oxidation state of terminal Br in Br_3O_8 is +7.

Statement II: Average oxidation number of Br in Br_3O_8 is $\frac{+16}{3}$.

In the light of above statements, choose the most appropriate answer from the options given below.

- (1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are correct
(4) Both statement I and statement II are incorrect

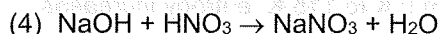
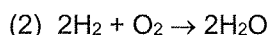
80. Number of compounds among the following in which halogen is in negative oxidation state is CsI_3 , NaBr , KClO_4 , Ca(OCI)_2 , O_2F_2

- (1) 2
(2) 4
(3) 5
(4) 3

Space for Rough Work



81. Disproportionation reaction among the following is,



82. Standard reduction potential of three metals in order of $E_{\text{A}^{2+}/\text{A}} > E_{\text{B}^{2+}/\text{B}} > E_{\text{C}^{2+}/\text{C}}$

Identify the correct statement among the following.

(1) A is soluble in BSO_4

(2) C is soluble in $\text{A}(\text{NO}_3)_2$

(3) $\text{C}^{2+}(\text{aq})$ is strongest oxidising agent

(4) $\text{B}(\text{s}) + \text{CSO}_4 \rightarrow \text{BSO}_4 + \text{C}(\text{s})$ will have positive value of EMF

83. Which among the following will have maximum limiting molar conductivity?

(1) $\text{Na}^+(\text{aq})$ (2) $\text{K}^+(\text{aq})$

(3) $\text{Mg}^{2+}(\text{aq})$ (4) $\text{H}^+(\text{aq})$

84. Standard cell potential for a cell represented as $\text{Cu}(\text{s})|\text{Cu}^{2+}(0.01\text{M})||\text{Cu}^{2+}(0.001\text{M})|\text{Cu}(\text{s})$ will be

(1) 0.029 V (2) -0.029 V

(3) 0.059 V (4) 0

85. Total volume of gas evolved at STP on electrolysis of 2 M $\text{CuSO}_4(\text{aq})$ using Pt electrodes by passing 1 F charge will be

(1) 16.8 L (2) 5.6 L

(3) 33.6 L (4) 44.8 L

SECTION - B

86. For a cell involving two electrons, $E_{\text{cell}}^\circ = 1.18 \text{ V}$ at 298 K, the equilibrium constant for cell reaction is

(1) 10^{10} (2) 10^{20}

(3) 10^{30} (4) 10^{40}

87. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Electrode potential increases with increase in size of metal rod.

Reason (R): Electrode potential is intensive property.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are true but R is not the correct explanation of A

(2) A is true but R is false

(3) A is false but R is true

(4) Both A and R are true and R is the correct explanation of A

88. Most reactive metal in aqueous medium is

(1) Li (2) Na

(3) Pb (4) Zn

89. The solubility of Hg_2Cl_2 in water is $S \text{ mol L}^{-1}$ at 298 K. The value of its solubility product K_{sp} will be

(1) S^2 (2) $16S^4$

(3) $27S^4$ (4) $4S^3$

90. If 18 kg of aluminium is deposited on passing certain amount of electricity through AlCl_3 solution, then mass of silver deposited from AgNO_3 by passing equal amount of electricity will be (Atomic mass of Ag = 108 u)

(1) 108 kg (2) 216 kg

(3) 54 kg (4) 162 kg

91. PCl_5 is sealed in a container, and at equilibrium 40% of volume is occupied by PCl_5 at 5 atm equilibrium pressure then K_p of reaction will be

(1) 2.25 atm (2) 4 atm

(3) 1.125 atm (4) 4.5 atm

92. Oxidation state of Cr in K_2CrO_4 is

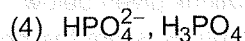
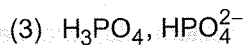
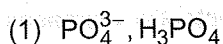
(1) +5 (2) +6

(3) +4 (4) +7

Space for Rough Work



93. Conjugate acid and base respectively of H_2PO_4^- are



94. Given below are two statements

Statement I: K_c increases with increase in temperature for endothermic reaction.

Statement II: K_c decreases with increase in temperature for exothermic reaction.

In the light of above statements, choose the most appropriate answer from the options given below

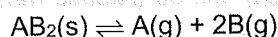
(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

95. For reaction



If $K_p = 108$, then total pressure at equilibrium state will be

(1) 6 atm (2) 3 atm

(3) 27 atm (4) 9 atm

96. For a reaction $\Delta H = 20 \text{ kJ mol}^{-1}$ and $\Delta S = 40 \text{ J mol}^{-1} \text{ K}^{-1}$ then equilibrium temperature for reaction will be

(1) 500°C (2) 1000°C

(3) 227°C (4) 727°C

97. Given below are two statements

Statement I: Magnitude of work done in adiabatic expansion is more than work done in isothermal expansion.

Statement II: If temperature of a crystalline solid is raised from 0 K to 115 K, entropy increases.

In the light of above statements, choose the most appropriate answer from the options given below

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

98. Which among the following acids will give maximum heat on complete neutralisation with aqueous NaOH?

(1) 1 mol HCl (2) 1 mol H_2SO_4

(3) 1 mol CH_3COOH (4) 1 mol HF

99. Given below are the two statements

Statement I: Species that loses electron is called reducing agent during a redox reaction.

Statement II: Species that gains electron is called oxidising agent during a redox reaction.

In the light of above statements, choose the most appropriate answer from the options given below

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

100. Which among the following species cannot get disproportionated?

(1) Cl_2 (2) MnO_2

(3) HOCl (4) H_2S

Space for Rough Work



[BOTANY]

SECTION - A

101. Casparian strips are present in the endodermis of
 (1) Dicot stem (2) Dicot root
 (3) Monocot stem (4) Dicot leaf
102. The epidermis in a dorsiventral leaf
 (1) Is not covered by cuticle
 (2) Covers both adaxial and abaxial surfaces
 (3) Is also known as epiblema
 (4) Lack living cells
103. Select the **correct** statement.
 (1) Mesophyll is differentiated into palisade and spongy parenchyma in grasses.
 (2) Vascular bundles are conjoint and closed in leaves.
 (3) The abaxial surface generally bear less stomata as compare to adaxial surface in dorsiventral leaves.
 (4) The size of vascular bundles are independent of the size of veins in dicot leaf.
104. Match the following columns and choose the **correct** option.

	Column I		Column II
a.	Xylem parenchyma	(i)	Absent in most of the monocots
b.	Sclereids	(ii)	Water transporting elements of xylem
c.	Phloem parenchyma	(iii)	Fruit walls of nuts
d.	Tracheids	(iv)	Store food in form of starch or fat and other substances like tannins

- (1) a(i), b(ii), c(iii), d(iv) (2) a(iv), b(iii), c(i), d(ii)
 (3) a(iii), b(ii), c(iv), d(i) (4) a(i), b(iii), c(ii), d(iv)

105. Read the following features (a-d) w.r.t. phloem fibres.
 a. Chlorenchymatous in nature
 b. Generally absent in the secondary phloem.
 c. Elongated, unbranched and have pointed, needle like apices.
 d. At maturity, these fibres lose their protoplasm and become dead.

The **incorrect** ones are

- (1) a and b (2) a and c
 (3) c and d (4) b and d
106. A few epidermal cells, in the vicinity of the guard cells become specialised in their shape and size and are known as
 (1) Bundle sheath cells
 (2) Subsidiary cells
 (3) Trichomes
 (4) Bulliform cells
107. In dicot root, initiation of vascular cambium during the secondary growth takes place in which cells?
 (1) Endodermis (2) Pericycle
 (3) Cortex (4) Epidermis
108. Read the following statements.
Statement A: Hypodermis consists of 3 to 5 layers of collenchymatous cells in dicot stem.
Statement B: All tissues except epidermis constitute the ground tissue.
 In the light of above statements choose the **correct** option.
 (1) Both statements are incorrect
 (2) Both statements are correct
 (3) Only statement A is correct
 (4) Only statement B is correct

Space for Rough Work



109. Which of the following have endodermis that are rich in starch grains?
- Dicot root
 - Monocot stem
 - Dicot stem
 - Monocot root
110. Outermost part of stele is
- Vascular bundles
 - Pith
 - Pericycle
 - Endodermis
111. Mark the **incorrect** statement w.r.t. guard cells of stomata.
- In grasses, they are dumb-bell shaped.
 - They possess chloroplasts.
 - Regulate the opening and closing of stomata.
 - Outer walls are thick and inner walls are thin in monocots.
112. Read the given statements w.r.t. meristematic tissues.
- A. The cells which are structurally and functionally specialised and lose the ability to divide, are called meristems.
- B. The meristems which occur at the tips of root and shoots contributes to the formation of the primary plant body.
- C. Lateral meristems are always secondary meristem.
- D. There are specific regions of apical meristem that are responsible for production of dermal tissues, ground tissues and vascular tissues.
- In light of above statements choose the **correct** option.
- Only A and B
 - Only A, B and C
 - Only D
 - Only B and D
113. In the dicot roots the lateral roots originates from
- Parenchyma between endodermis and pericycle
 - Innermost region of the stele
 - Parenchymatous cells present next to endodermis
 - Intrafascicular and interfascicular tissue in a ring
114. The permanent tissue that forms the major component within the various organs of plants
- Is thin-walled and performs the function of photosynthesis, storage and secretion
 - Lacks intercellular space and provides mechanical support to the growing parts of the plant
 - Is usually dead without protoplast and has thick and lignified cell walls
 - Have thickened cells at corners which is due to presence of cellulose, hemicellulose and pectin
115. Radial conduction of water takes place by the
- Phloem parenchyma
 - Xylem fibres
 - Ray parenchyma
 - Vessels
116. Monocot root is anatomically similar to the dicot root in
- Presence of well developed and large central pith
 - Having diarch to hexarch vascular bundles
 - Possessing radial vascular bundles with exarch protoxylem
 - Absence of cambium in the vascular bundle throughout the life
117. Select the **correct** option for monocotyledonous stem.
- Water-containing cavities are found in pith
 - Vascular bundle is surrounded by a sheath made up of sclerenchymatous tissue
 - Well developed medullary rays are present
 - Peripheral vascular bundles are generally larger than centrally located ones.
118. Natural system of classification was given by
- Linnaeus
 - Bentham and Hooker
 - Aristotle
 - Engler and Prantl

Space for Rough Work



119. Gametes are flagellated and similar in size in

- (1) *Spirogyra* (2) *Eudorina*
(3) *Ulothrix* (4) *Fucus*

120. Select the **incorrect** match.

- (1) *Polysiphonia* – Oogamous reproduction with motile male gametes
(2) *Spirogyra* – Filamentous algae
(3) *Laminaria* – Brown algae and rich source of iodine
(4) *Ectocarpus* – Body is simple branched and filamentous

121. Which of the following photosynthetic organism are rich in proteins and are used as food supplements even by space travellers?

- (1) *Volvox* and *Chlorella*
(2) *Gelidium* and *Porphyra*
(3) *Chlorella* and *Spirulina*
(4) *Chara* and *Polysiphonia*

122. Which of the following characters is specific to red algae?

- (1) Cell wall is made up of an inner layer of cellulose and an outer layer of pectose
(2) Complex post-fertilisation development is seen
(3) Presence of fucoxanthin and holdfast
(4) Pear-shaped gametes with two laterally attached flagella

123. Read the following statements for bryophytes and identify them as **true (T)** or **false (F)**.

- I. Gametophyte is the haploid stage of the plant that generates gametes by the process of mitosis.
II. Gametophyte of bryophyte consist of multicellular sex organs.
III. Gemmae are non-green, unicellular, asexual buds.
IV. They are non-vascular and are the first embryophytes.

Select the **correct** option.

- | | I | II | III | IV |
|-----|---|----|-----|----|
| (1) | T | F | T | F |
| (2) | F | T | F | T |
| (3) | T | T | F | T |
| (4) | F | T | T | F |

124. Gymnosperms are adapted to tolerate extreme environmental conditions because of

- (1) Large surface area of leaves
(2) Thin cuticle on leaf surface
(3) Sunken stomata
(4) Fan shaped leaves

125. Read the following statements (A-E).

- A. Vegetative reproduction in mosses is by fragmentation and budding in the secondary protonema.
B. In liverworts, mosses and gymnosperm gametophytes are free-living.
C. The leaves in pteridophyte are microphylls as in ferns or macrophylls as in *Selaginella*.
D. The sporophyte in mosses is more elaborate than that in liverworts.
E. *Marchantia* is a monoecious liverwort.

How many of the above statements are **incorrect**?

- (1) One (2) Three
(3) Two (4) Four

126. In *Selaginella*, seed formation does not take place because

- (1) Integumented microsporangium is absent
(2) Development of the zygotes into young embryos take place within the female gametophytes
(3) Megaspores are not retained permanently within the megasporangium
(4) Heterosporous condition is absent

127. Peat is obtained from

- (1) *Marchantia* (2) *Polytrichum*
(3) *Funaria* (4) *Sphagnum*

Space for Rough Work



128. Mark the **correct** match

- (1) Psilopsida – *Psilotum*
- (2) Sphenopsida – *Selaginella*
- (3) Pteropsida – *Equisetum*
- (4) Lycopsidea – *Dryopteris*

129. Which of the following is a common characteristic between pteridophytes and bryophytes?

- (1) Dominant sporophytic phase in the life cycle
- (2) Sporophyte is differentiated into foot, seta and capsule.
- (3) Presence of heterospory
- (4) Spores germinate to give rise to free-living, photosynthetic gametophyte

130. Read the statements carefully and mark the **correct** option.

- A. Pteridophytes are the first tracheophytes.
- B. In pteridophytes, gametophyte bear male and female sex organs called antheridia and archegonia respectively
- C. In some pteridophytes like *Salvinia*, the sporangia produce spores by mitosis that give rise to prothallus.

- (1) Only A is correct
- (2) Only C is correct
- (3) A and B are correct
- (4) A and C are correct

131. Gymnosperms are called naked seeded plants due to the absence of

- (1) Tracheids
- (2) Archegonia
- (3) Integumented megasporangia
- (4) Ovary wall

132. Prothallus in *Dryopteris* is

- a. Mostly photosynthetic
- b. Multicellular

- c. Large, conspicuous and dependent on sporophyte
- d. Haploid
- e. Thaloid gametophyte

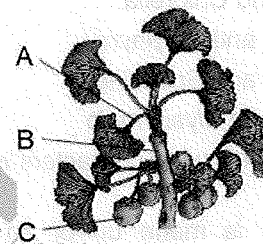
Select the **correct** option

- (1) Only a, b and c
- (2) Only b, c, d and e
- (3) Only a, b, d and e
- (4) Only c, d and e

133. Which of the following is an unicellular algae?

- (1) *Spirogyra*
- (2) *Ulothrix*
- (3) *Chlamydomonas*
- (4) *Laminaria*

134. Which of the following is **incorrect** w.r.t given figure?



- (1) A is dwarf and B is long shoot
- (2) Given organism is heterosporous
- (3) C represents the fruit
- (4) Given organism shows the presence of fan-shaped leaves

135. How many feature(s) is/are common between bryophytes and pteridophytes?

- a. True leaves
- b. Diploid sporophyte
- c. Jacketed sex organs
- d. Motile male gametes
- e. Are embryophytes

Select the **correct** option.

- (1) Three
- (2) Four
- (3) Two
- (4) One

Space for Rough Work



SECTION - B

136. Which of the following is **not** responsible for the formation of secondary tissues?

- (1) Interfascicular cambium
- (2) Intercalary meristem
- (3) Cork cambium
- (4) Intrafascicular cambium

137. The outermost layer in transverse section of dicot root is

- (1) Cortex
- (2) Endodermis
- (3) Pericycle
- (4) Epiblema

138. Read the following **Assertion (A)** and **Reason (R)** statements and choose the **correct** option.

Assertion (A): Bulliform cells are present in upper epidermis of grass leaves.

Reason (R): Bulliform cells make the leaves curl inward to minimise water loss during water stress.

- (1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (2) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (3) Both (A) and (R) are false
- (4) (A) is true but (R) is false

139. Select the **incorrect** statement from the following.

- (1) Trichomes help in preventing water loss due to transpiration
- (2) Cuticle prevents loss of water
- (3) Cuticle is absent in roots
- (4) Epidermis is made up of elongated, loosely arranged cells with large vacuole

140. Read the following statements and identify them as **true (T)** or **false (F)**

- A. Epidermis is usually single layered.
- B. Cortical layers below hypodermis of dicot stem consist of rounded thin walled parenchymatous cells with conspicuous intercellular spaces.

C. Root hairs are unicellular elongation of the epidermal cells.

Select the **correct** option.

- (1) A(T), B(F), C(T)
- (2) A(F), B(T), C(T)
- (3) A(T), B(T), C(T)
- (4) A(T), B(F), C(F)

141. The 'ring' arrangement of vascular bundles is a characteristic of

- (1) Monocot stem
- (2) Dicot stem
- (3) Monocot root
- (4) Dicot root

142. Which of the following is **not** true regarding companion cells?

- (1) Helps in maintaining the pressure gradient in the sieve tubes
- (2) Possesses a peripheral cytoplasm and a large vacuole but lacks a nucleus
- (3) Closely associated with sieve tube element
- (4) These are specialised parenchymatous cells

143. Which of the following statement is **true**?

- (1) Presence of vessels is a characteristic feature of angiosperms.
- (2) Interfascicular and cork cambium are redifferentiated cells.
- (3) A mature sieve tube element possesses conspicuous nucleus.
- (4) Each stoma is composed of two types of cells i.e., guard cell and subsidiary cells.

144. Karyotaxonomy is based on all **except**

- (1) Chromosome number
- (2) Chromosome structure
- (3) Aromatic compounds
- (4) Chromosome behaviour

Space for Rough Work



145. Pollen grains in gymnosperms represents
- Reduced endosperm
 - Microsporangium
 - Reduced male gametophyte
 - Male cone

146. Read the following statements and select the **correct** option.

Statement A: *Cycas* have small specialised roots called coralloid roots and are associated with N_2 -fixing cyanobacteria.

Statement B: In *Cycas*, stems are branched and the leaves are pinnately compound which persist for a few years.

- Both statement A and statement B are correct
- Only statement A is incorrect
- Both statement A and statement B are incorrect
- Only statement B is incorrect

147. Read the following **Assertion (A)** and **Reason (R)** statements and choose the **correct** option.

Assertion (A): Agar is used to grow microbes and in preparation of ice-creams and jellies.

Reason (R): Commercial source of agar is *Gelidium* and *Gracilaria*.

- Both (A) and (R) are true but (R) is not the correct explanation of (A)
- Both (A) and (R) are true and (R) is the correct explanation of (A)
- Both (A) and (R) are false
- (A) is true but (R) is false

148. Which of the following algal pairs reproduce sexually by non-motile gametes?

- Ectocarpus* and *Sargassum*
- Dictyota* and *Ulothrix*
- Gracilaria* and *Porphyra*
- Gelidium* and Kelps

149. Winged pollen grains are present in

- Mustard
- Cycas*
- Pinus*
- Wheat

150. Match the column I with column II and choose the **correct** option.

	Column I		Column II
a.	<i>Cycas</i>	(i)	Male or female cones are borne on same tree
b.	<i>Dictyota</i>	(ii)	Is a giant red wood tree
c.	<i>Sequoia</i>	(iii)	Presence of laminarin and fucoxanthin
d.	<i>Pinus</i>	(iv)	Male cones and megasporophylls are borne on different trees

- a(i), b(ii), c(iii), d(iv)
- a(iv), b(iii), c(ii), d(i)
- a(i), b(iv), c(iii), d(ii)
- a(iv), b(ii), c(iii), d(i)

[ZOOLOGY]

SECTION - A

151. In humans, a U-shaped bone present at the base of the buccal cavity is called

- Hyoid
- Sphenoid
- Lacrimal
- Maxilla

152. In *Periplaneta americana*, the function served by Malpighian tubules in addition to excretion is

- Balancing
- Respiration
- Defense
- Osmoregulation

Space for Rough Work



153. Parasympathetic neural system is a component of
- Autonomic neural system
 - Somatic neural system
 - Peripheral and somatic neural system
 - Central neural system
154. In the common form of chemical synapse, the neurotransmitters released in synaptic cleft bind to the specific receptors present on the
- Pre-synaptic membrane
 - Axonal membrane
 - Post-synaptic membrane
 - Synaptic vesicle
155. All of the following acts as a vasoconstrictor, **except**
- ADH
 - ANF
 - Vasopressin
 - Angiotensin II
156. In human brain, the layer of cranial meninges which is in direct contact with the brain tissue is
- Grey matter
 - Arachnoid
 - Pia mater
 - Dura mater
157. Flagellar movements help in all of the following activities, **except**
- Maintenance of water current in the canal system of *Sycon*
 - Swimming of human spermatozoa
 - Locomotion of *Euglena*
 - Movement of food through cytopharynx in *Paramoecium*
158. Consider the statements A and B given below.
- Statement A:** All locomotions are movements but all movements are not locomotions.
- Statement B:** Macrophages in blood and monocytes in tissue exhibit amoeboid movements.
- Select the **correct** option.
- Both statements A and B are correct
 - Both statements A and B are incorrect
 - Only statement A is correct
 - Only statement B is correct
159. Choose the **incorrect** option w.r.t. electrical synapses observed in humans.
- Transmission of an impulse across it is similar to impulse conduction along a single axon
 - Pre and post-synaptic membranes of neurons are in close proximity
 - Present abundantly
 - Neurotransmitters are not involved in the conduction of impulses across it
160. Each organised skeletal muscle in the human body is made of a number of _____ held together by a common collagenous connective tissue layer called _____.
- Choose the **correct** option to fill the respective blanks.
- Fascicles; Fascia
 - Fascicles; Sarcolemma
 - Fascia; Fascicles
 - Sarcolemma; Fascia
161. How many bones given in the box below is/are part of appendicular skeleton in a normal human body?
- Patella, Vertebrae, Clavicle, Scapula
- Select the **correct** option.
- One
 - Two
 - Three
 - Four
162. During the process of tubular reabsorption in humans, all of the following substances are reabsorbed actively, **except**
- Glucose
 - Amino acids
 - Na⁺
 - Urea

Space for Rough Work



163. The $\text{Na}^+ - \text{K}^+$ pump which maintains ionic gradients across the resting neuronal membrane, transports

- (1) 2 Na^+ inwards for 3 K^+ out of the cell
- (2) 2 K^+ outwards for 3 Na^+ into the cell
- (3) 3 Na^+ outwards for 2 K^+ into the cell
- (4) 3 K^+ inwards for 2 Na^+ out of the cell

164. The anatomical unit of muscle is 'X' whereas 'Y' is its functional unit. Identify 'X' and 'Y' and select the correct option.

	'X'	'Y'
(1)	Sarcomere	Muscle fibre
(2)	Muscle fibre	Sarcomere
(3)	Myofibrils	Muscle bundle
(4)	Sarcomere	Myofibril

165. In humans, the large triangular flat bone of the pectoral girdle, that is situated in the dorsal part of the thorax between the second and the seventh ribs is

- (1) Acromion
- (2) Scapula
- (3) Clavicle
- (4) Pubis

166. Select the **incorrect** statement w.r.t. excretory system of humans.

- (1) The osmolarity gradient in the cortex and medullary interstitium of kidney is mainly caused by NaCl and urea.
- (2) Osmolarity of about $1200 \text{ mOsmol L}^{-1}$ is present in the inner medullary interstitium of kidney.
- (3) The proximity between Henle's loop and vasa recta, contribute in increasing the osmolarity at the lower end of Henle's loop.
- (4) The flow of filtrate in the two limbs of Henle's loop is in the same direction.

167. During the resting state, the axonal membrane of a neuron is

- (1) Comparatively less permeable to K^+ than Na^+
- (2) In a depolarised state
- (3) Nearly impermeable to Na^+
- (4) Permeable to negatively charged proteins

168. The part of human brain that lies at the base of the thalamus contains neurosecretory cells and also has control centres for

- (1) Cardiovascular reflexes
- (2) Respiration
- (3) Maintaining balance and posture of the body
- (4) Thermoregulation

169. In humans, the hormone that keeps a check on the renin-angiotensin mechanism is released from the

- (1) JG cells of kidney
- (2) Adrenal cortex
- (3) Neurohypophysis
- (4) Atria of the heart

170. Read the given statements.

- a. Neurons can detect, receive, produce and transmit stimuli.
- b. Green glands perform excretion in *Amphioxus*.
- c. In human kidneys, the cortex extends in between the medullary pyramids as renal columns called columns of Bertini.
- d. On an average, 15 – 20 gm of urea is excreted out through urine per day by a healthy man.

How many of them is/are **correct**?

- (1) One
- (2) Two
- (3) Three
- (4) Four

171. In humans, the part of brain that is connected to spinal cord and has centres which control gastric secretions is

- (1) Cerebellum
- (2) Medulla oblongata
- (3) Thalamus
- (4) Hypothalamus

Space for Rough Work



172. Select the option that represents an **incorrect** difference between the red and white muscle fibres.

	Features	Red muscle fibres	White muscle fibres
(1)	Content of myoglobin	High	Low
(2)	Amount of mitochondria	Plenty	Few
(3)	Amount of sarcoplasmic reticulum	Low	High
(4)	Major energy source	By anaerobic process	By aerobic process

173. In humans, juxtaglomerular apparatus, a special sensitive region, is formed by the cellular modifications of the part of renal tubule where _____ and the _____ at the location of their contact.

Select the **correct** option to fill in the respective blanks.

- (1) Conditional reabsorption of Na^+ and water takes place; afferent arteriole
- (2) Maximum reabsorption of electrolytes takes place; afferent arteriole
- (3) Conditional reabsorption of water and Na^+ takes place; efferent arteriole
- (4) Maximum reabsorption of nutrients takes place; efferent arteriole

174. **Assertion (A):** Vertebrochondral ribs are called false ribs.

Reason (R): These ribs do not articulate directly with the sternum but join the seventh rib with the help of hyaline cartilage.

In the light of above statements, select the **correct** option w.r.t. humans.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

175. Consider an axonal membrane in which a segment gets depolarised. Now, with respect to this segment, the direction of flow of current on the inner surface and on the outer surface of axonal membrane is in which direction?

- (1) Same
- (2) Opposite
- (3) Perpendicular
- (4) Parallel

176. Match column I with column II w.r.t. types of neurons in humans and their respective locations.

	Column I		Column II
a.	Unipolar	(i)	Cerebral cortex
b.	Bipolar	(ii)	Retina of eye
c.	Multipolar	(iii)	Embryonic stage

Select the **correct** option.

- (1) a(i), b(ii), c(iii)
- (2) a(iii), b(i), c(ii)
- (3) a(i), b(iii), c(ii)
- (4) a(iii), b(ii), c(i)

177. At the fusion point of ilium, ischium and pubis, is a cavity to which _____ articulates.

Choose the **correct** option to fill in the blank.

- (1) Acetabulum
- (2) Pubic symphysis
- (3) Coxal bone
- (4) Femur

Space for Rough Work



178. Read the given statements w.r.t. humans.

Statement A: Glomerulus is not a part of renal tubule.

Statement B: Loop of Henle lies mainly in the medullary region of kidneys.

Select the **correct** option.

- (1) Both statements A and B are correct
- (2) Both statements A and B are incorrect
- (3) Only statement A is correct
- (4) Only statement B is correct

179. Brain stem forms the connections between brain and spinal cord. Brain stem includes three major regions which are

- (1) Cerebrum, cerebellum and thalamus
- (2) Hypothalamus, mid brain and thalamus
- (3) Midbrain, pons and medulla oblongata
- (4) Pons, cerebellum and medulla oblongata

180. Consider the following statements w.r.t. an adult human.

- a. Normally, kidneys produce urine that is hypotonic than the initial filtrate formed.
- b. Renin converts angiotensinogen in blood to angiotensin I.
- c. Angiotensin II activates the adrenal medulla to release aldosterone.
- d. The total weight of both the kidneys is 120 – 170 g.

How many of the above statements is/are **incorrect**?

- (1) One
- (2) Two
- (3) Three
- (4) Four

181. In a nephron, podocytes are the intricately arranged specialised cells of

- (1) Epithelium of afferent arteriole
- (2) Endothelium of efferent arteriole
- (3) Endothelium of glomerular blood vessel
- (4) Epithelium of Bowman's capsule

182. In humans, sebaceous glands through sebum majorly eliminate

- (1) Lactic acid
- (2) Waxes
- (3) Bilirubin
- (4) Urea

183. The type of muscle fibres associated with biceps differ from smooth muscle fibres in all of the following features, **except**

- (1) Location of nuclei
- (2) Shape
- (3) Presence of striations
- (4) Absence of branching

184. **Assertion (A):** Skeletal muscle fibres appear striped when observed under the microscope.

Reason (R): Each myofibril in skeletal muscles has alternate actin and myosin filaments.

In the light of above statements, select the **correct** option.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

185. The (A) portion of (B) consists mainly of four round swellings called corpora quadrigemina. Identify (A) and (B) and select the **correct** option w.r.t. humans.

- (1) (A) – Dorsal, (B) – Midbrain
- (2) (A) – Ventral, (B) – Hindbrain
- (3) (A) – Dorsal, (B) – Hindbrain
- (4) (A) – Ventral, (B) – Midbrain

Space for Rough Work



SECTION - B

186. Choose the **odd** one w.r.t. the functions of association areas present in the human brain.
- (1) Intersensory association
 - (2) Memory
 - (3) Communication
 - (4) Maintaining balance and posture of the body
187. In humans, the cerebrum is divided into left and right cerebral hemispheres by
- (1) Corpus callosum
 - (2) A tract of nerve fibres
 - (3) A deep cleft
 - (4) Cerebral aqueduct
188. In humans, during the release of urine
- (1) Urethral sphincters relax
 - (2) Smooth muscles of urinary bladder relax
 - (3) Urethral sphincters contract
 - (4) Skeletal muscles of urinary bladder relax
189. For regulation of body fluid volume, during dehydration, vasoconstriction occurs that leads to
- (1) Increase in blood pressure
 - (2) Decrease in GFR
 - (3) Decrease in blood pressure
 - (4) Increase in production of diluted urine
190. Choose the correct set of contractile and regulatory proteins respectively that participate in skeletal muscle contraction of humans.
- (1) Tropomyosin; Actin
 - (2) Actin; Meromyosin
 - (3) Myosin; Troponin
 - (4) Tropomyosin; Troponin
191. The amount of CO_2 normally removed by human lungs per minute is
- | | |
|------------|------------|
| (1) 200 mL | (2) 150 mL |
| (3) 100 mL | (4) 50 mL |
192. Choose the **incorrect** match w.r.t. adult humans.
- | | |
|-------------------------------|------|
| (1) Carpals in both forelimbs | – 16 |
| (2) Tarsals in both hindlimbs | – 14 |
| (3) Thoracic vertebrae | – 12 |
| (4) Vertebrosteral ribs | – 7 |
193. Under normal conditions, the neural system that relays impulses from CNS to triceps is
- (1) Somatic neural system
 - (2) Autonomic neural system
 - (3) Sympathetic neural system
 - (4) Visceral neural system
194. When a stimulus is applied, the electrical potential difference which develops across the plasma membrane of a neuron is termed as
- | | |
|-----------------------|-------------------|
| (1) Repolarisation | (2) Nerve impulse |
| (3) Resting potential | (4) Synapse |
195. Listed below are some functions
- a. Performs selective secretion of H^+
 - b. Performs selective secretion of K^+
 - c. Allows passage of small amounts of urea in medullary interstitium
- Select the part of kidney that can be associated with all of the above mentioned functions.
- (1) PCT
 - (2) DCT
 - (3) Collecting duct
 - (4) Glomerulus
196. In humans, the type of joint present between the vertebra that articulates with the occipital condyles and the vertebra next to it, is
- (1) Fibrous joint
 - (2) Saddle joint
 - (3) Gliding joint
 - (4) Pivot joint

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197. Match the columns w.r.t. disorders of muscular and skeletal system in humans.

	Column I		Column II
a.	Myasthenia gravis	(i)	Rapid spasms in muscles due to low Ca^{+2} levels in body fluid
b.	Muscular dystrophy	(ii)	Auto-immune disorder
c.	Tetany	(iii)	Accumulation of uric acid crystals in joints
d.	Gout	(iv)	Genetic disorder

Select the **correct** option.

- (1) a(iv), b(ii), c(iii), d(i) (2) a(iv), b(ii), c(i), d(iii)
(3) a(ii), b(iv), c(i), d(iii) (4) a(ii), b(iv), c(iii), d(i)

198. Which of the following holds true when the skeletal muscle is in a fully contracted state?

- (1) H-zone disappears completely
(2) Length of sarcomere remains the same
(3) Length of A-band decreases
(4) Length of I-band increases

199. Arrange the following organisms in the correct order w.r.t. the level of toxicity of the chief nitrogenous wastes that they excrete.

- A. Bony fishes
B. Aquatic amphibians
C. Terrestrial amphibians
D. Reptiles

Select the **correct** option.

- (1) $A < B < C = D$ (2) $C > D = A > B$
(3) $A = B > C > D$ (4) $D > B > C = A$

200. **Assertion (A):** Protein synthesising granules, called the Nissl's granules, are absent in axons of neurons.

Reason (R): Nissl's granules are composed of RER and ribosomes.

In the light of above statements, choose the **correct** option.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true, (R) is false
(4) Both (A) and (R) are false

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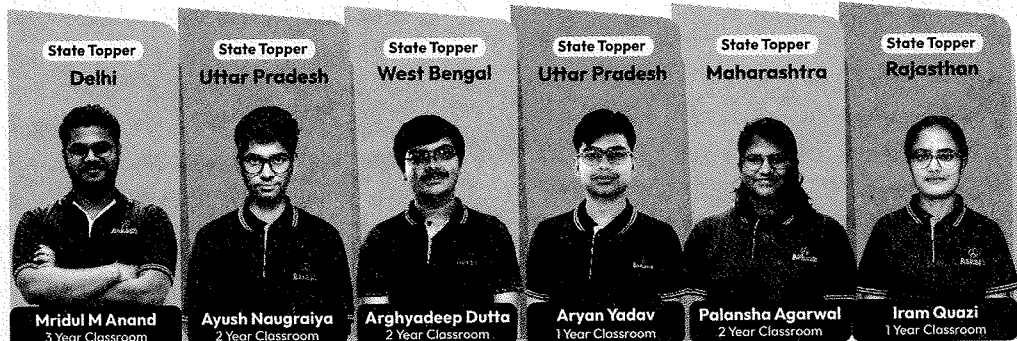


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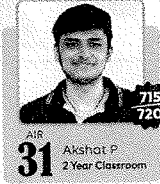
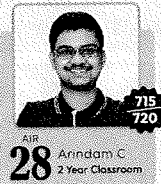
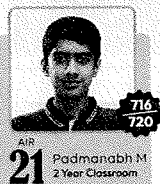
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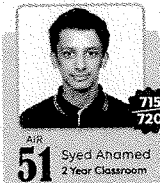
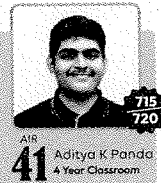
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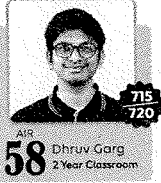
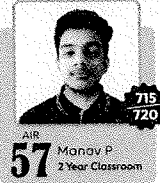
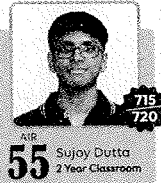
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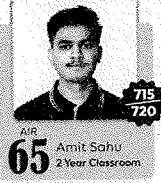
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